

Database Systems: Design, Implementation, and Management

Eighth Edition

Chapter 12

Distributed Database Management Systems

Objectives

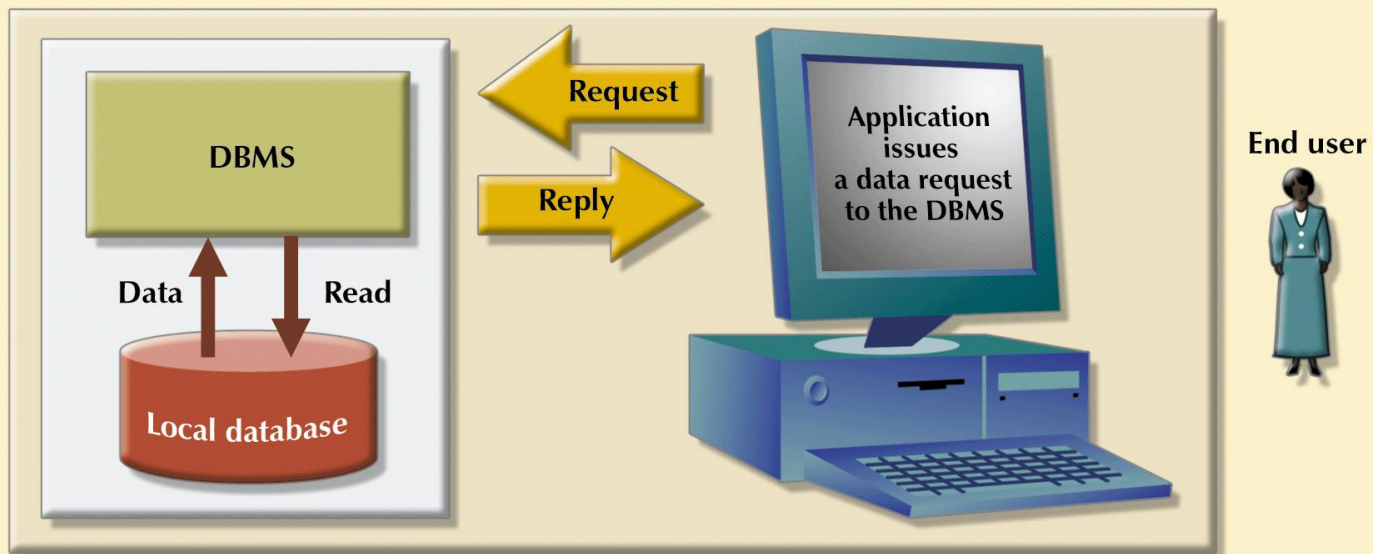
- In this chapter, you will learn:
 - What a distributed database management system (DDBMS) is and what its components are
 - How database implementation is affected by different levels of data and process distribution
 - How transactions are managed in a distributed database environment
 - How database design is affected by the distributed database environment

The Evolution of Distributed Database Management Systems

- **Distributed database management system (DDBMS)**
 - Governs storage and processing of logically related data
 - Interconnected computer systems
 - Both data and processing functions are distributed among several sites
- Centralized database required that corporate data be stored in a single central site

**FIGURE
12.1**

Centralized database management system



DDBMS Advantages and Disadvantages

- Advantages:
 - Data are located near “greatest demand” site
 - Faster data access
 - Faster data processing
 - Growth facilitation
 - Improved communications
 - Reduced operating costs
 - User-friendly interface
 - Less danger of a single-point failure
 - Processor independence

DDBMS Advantages and Disadvantages (continued)

- Disadvantages:
 - Complexity of management and control
 - Security
 - Lack of standards
 - Increased storage requirements
 - Increased training cost

**TABLE
12.1**

Distributed DBMS Advantages and Disadvantages

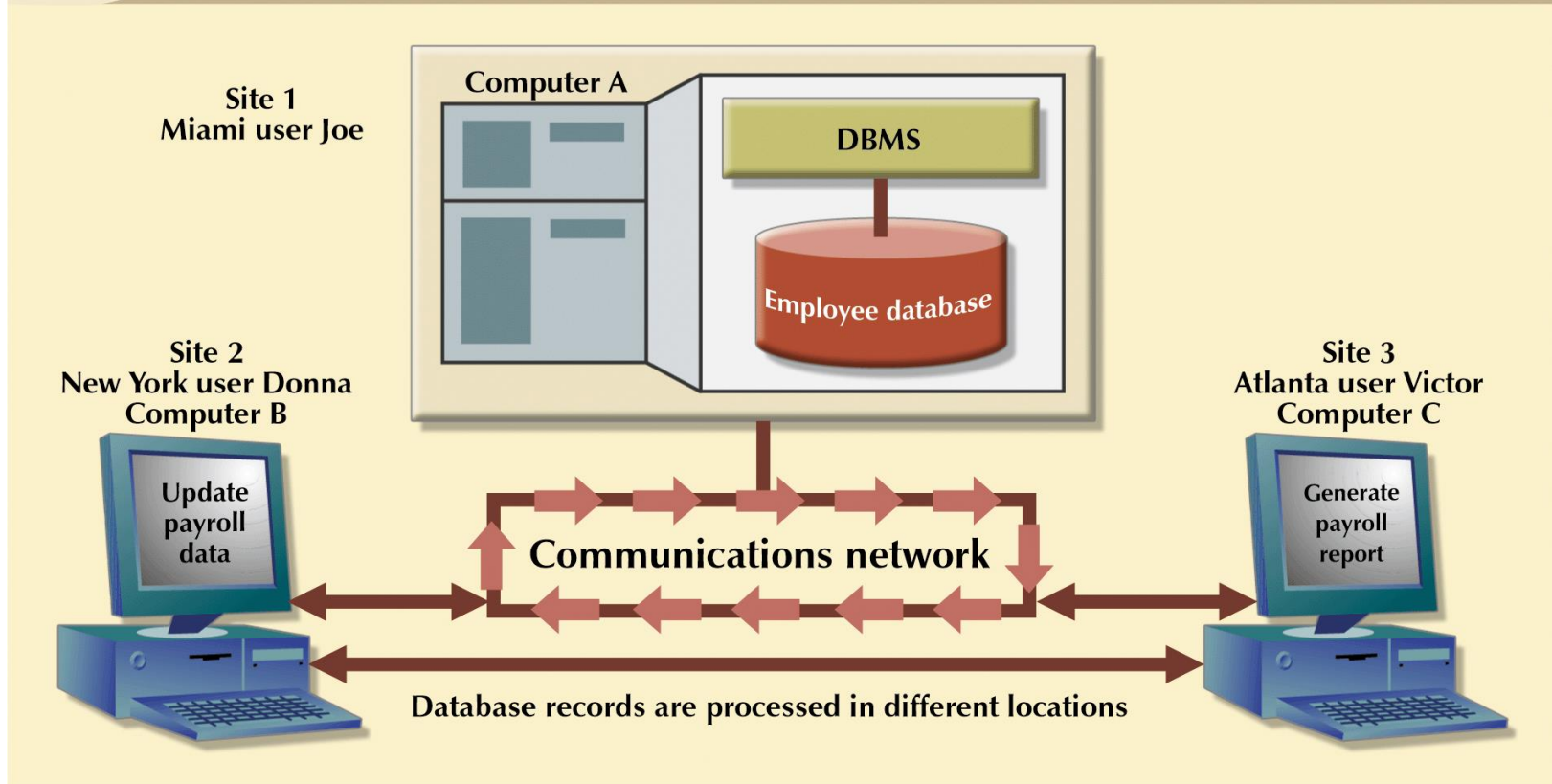
ADVANTAGES	DISADVANTAGES
• <i>Data are located near the greatest demand site.</i> The data in a distributed database system are dispersed to match business requirements. • <i>Faster data access.</i> End users often work with only a locally stored subset of the company's data. • <i>Faster data processing.</i> A distributed database system spreads out the systems workload by processing data at several sites. • <i>Growth facilitation.</i> New sites can be added to the network without affecting the operations of other sites. • <i>Improved communications.</i> Because local sites are smaller and located closer to customers, local sites foster better communication among departments and between customers and company staff. • <i>Reduced operating costs.</i> It is more cost-effective to add workstations to a network than to update a mainframe system. Development work is done more cheaply and more quickly on low-cost PCs than on mainframes. • <i>User-friendly interface.</i> PCs and workstations are usually equipped with an easy-to-use graphical user interface (GUI). The GUI simplifies training and use for end users. • <i>Less danger of a single-point failure.</i> When one of the computers fails, the workload is picked up by other workstations. Data are also distributed at multiple sites. • <i>Processor independence.</i> The end user is able to access any available copy of the data, and an end user's request is processed by any processor at the data location.	• <i>Complexity of management and control.</i> Applications must recognize data location, and they must be able to stitch together data from various sites. Database administrators must have the ability to coordinate database activities to prevent database degradation due to data anomalies. • <i>Technological difficulty.</i> Data integrity, transaction management, concurrency control, security, backup, recovery, query optimization, access path selection, and so on, must all be addressed and resolved. • <i>Security.</i> The probability of security lapses increases when data are located at multiple sites. The responsibility of data management will be shared by different people at several sites. • <i>Lack of standards.</i> There are no standard communication protocols at the database level. (Although TCP/IP is the de facto standard at the network level, there is no standard at the application level.) For example, different database vendors employ different—and often incompatible—techniques to manage the distribution of data and processing in a DDBMS environment. • <i>Increased storage and infrastructure requirements.</i> Multiple copies of data are required at different sites, thus requiring additional disk storage space. • <i>Increased training cost.</i> Training costs are generally higher in a distributed model than they would be in a centralized model, sometimes even to the extent of offsetting operational and hardware savings. • <i>Costs.</i> Distributed databases require duplicated infrastructure to operate (physical location, environment, personnel, software, licensing, etc.)

Distributed Processing and Distributed Databases

- **Distributed processing**
 - Database's logical processing is shared among two or more physically independent sites
 - Connected through a network
- **Distributed database**
 - Stores logically related database over two or more physically independent sites
 - Database composed of **database fragments**

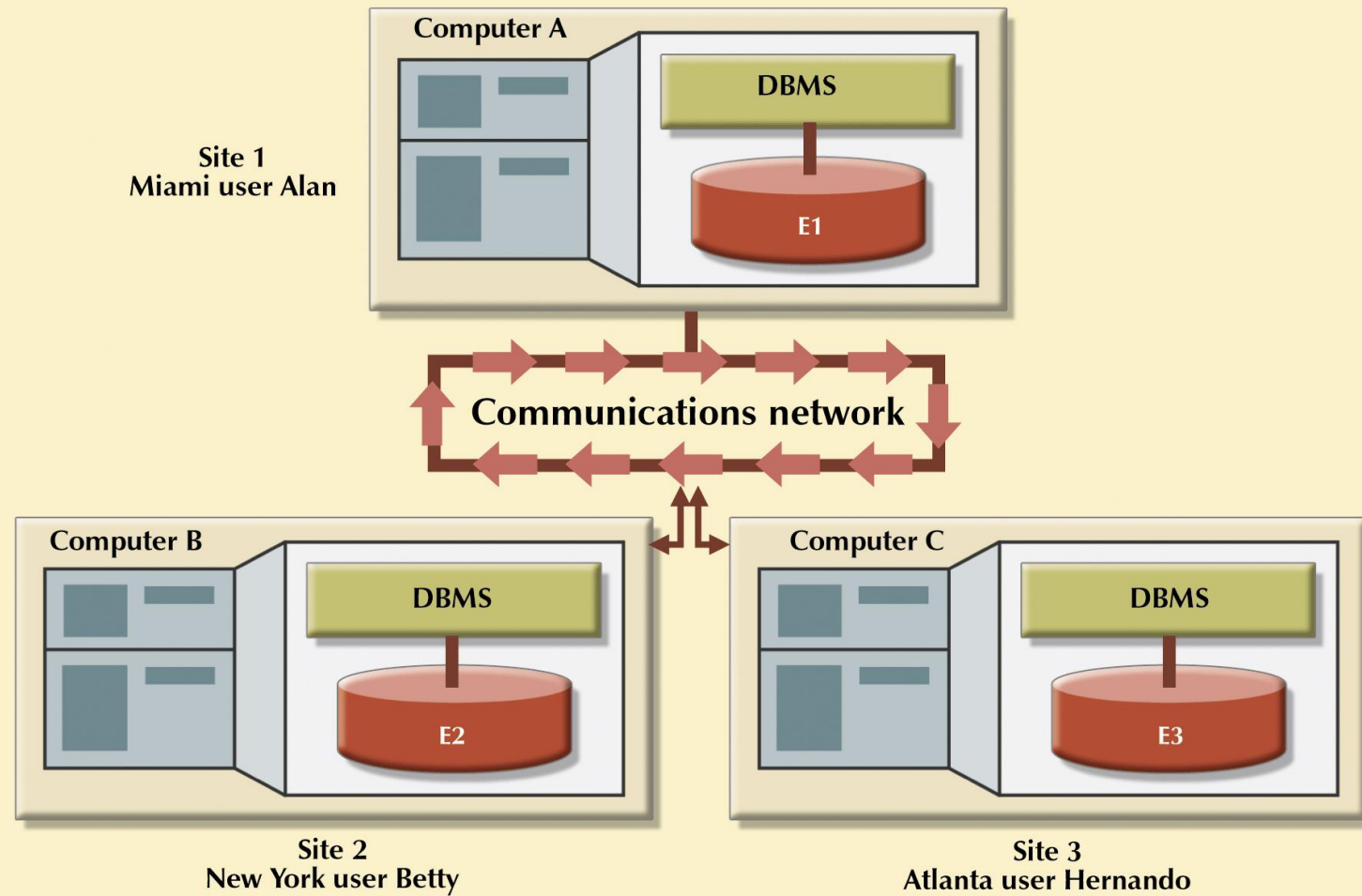
**FIGURE
12.2**

Distributed processing environment



**FIGURE
12.3**

Distributed database environment



Characteristics of Distributed Management Systems

- Application interface
- Validation
- Transformation
- Query optimization
- Mapping
- I/O interface

Characteristics of Distributed Management Systems (continued)

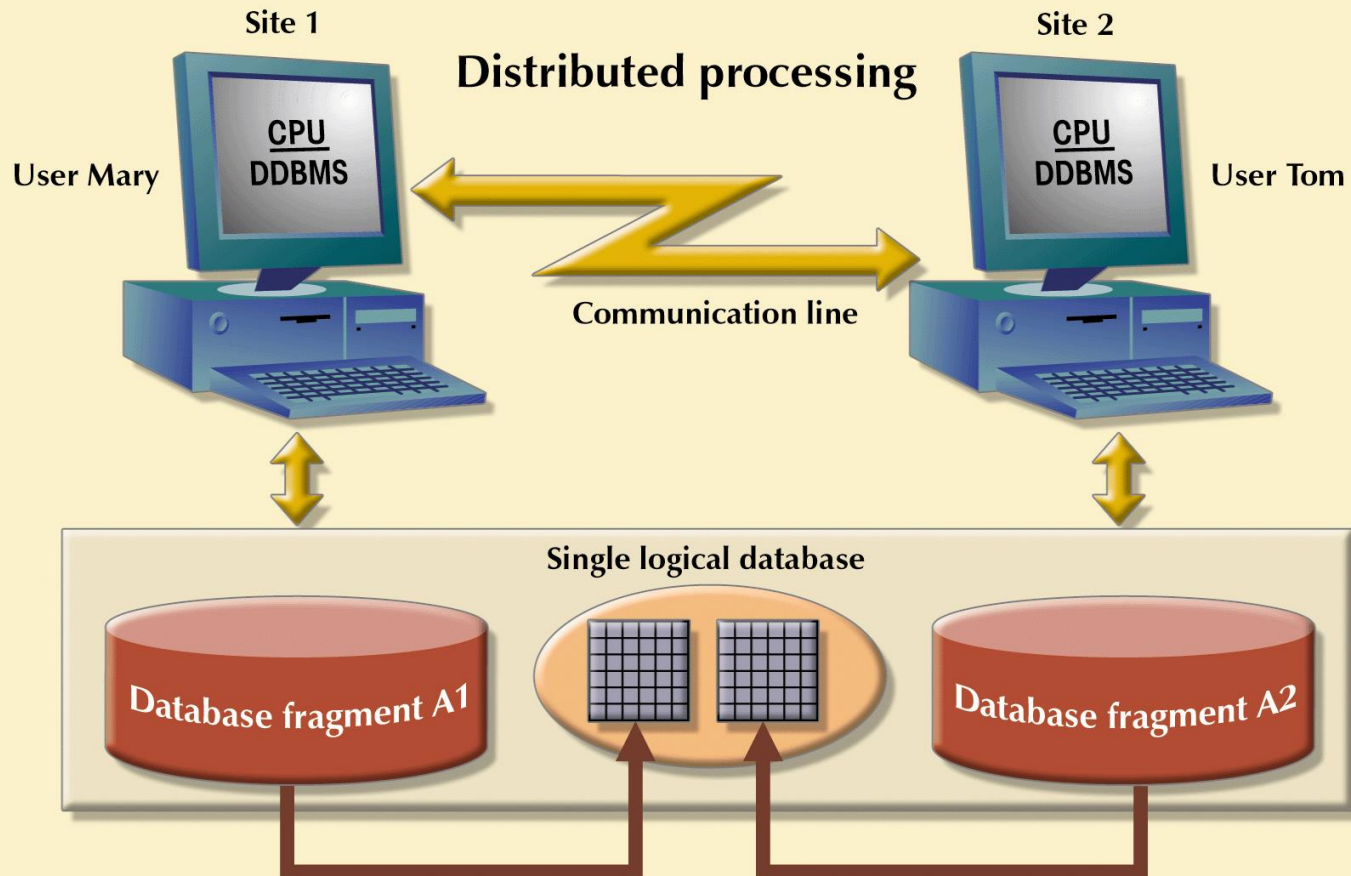
- Formatting
- Security
- Backup and recovery
- DB administration
- Concurrency control
- Transaction management

Characteristics of Distributed Management Systems (continued)

- Must perform all the functions of centralized DBMS
- Must handle all necessary functions imposed by distribution of data and processing
 - Must perform these additional functions transparently to the end user

**FIGURE
12.4**

A fully distributed database management system



DDBMS Components

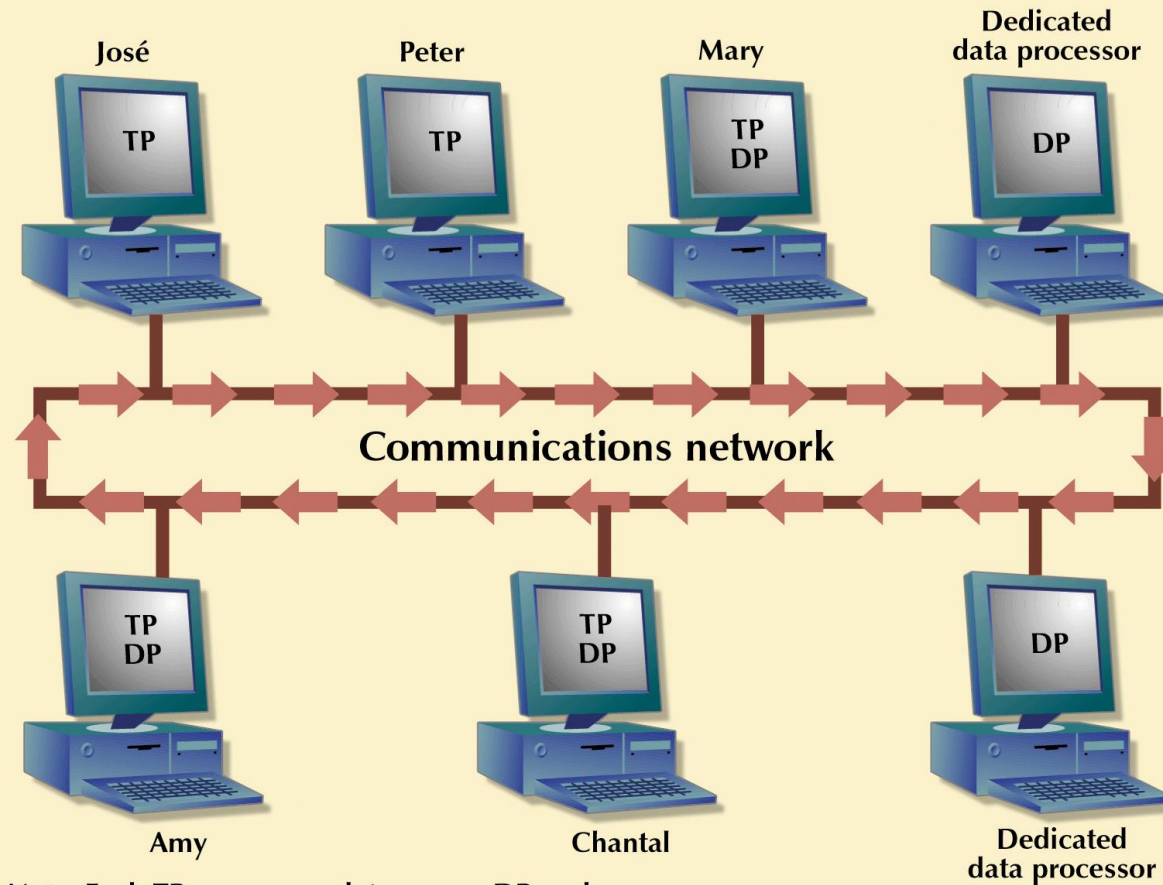
- Must include (at least) the following components:
 - Computer workstations
 - Network hardware and software
 - Communications media
 - **Transaction processor (application processor, transaction manager)**
 - Software component found in each computer that requests data

DDBMS Components (continued)

- Must include (at least) the following components: (continued)
 - **Data processor or data manager**
 - Software component residing on each computer that stores and retrieves data located at the site
 - May be a centralized DBMS

**FIGURE
12.5**

Distributed database system management components



Note: Each TP can access data on any DP, and each DP handles all requests for local data from any TP.

Levels of Data and Process Distribution

- Current systems classified by how process distribution and data distribution supported

TABLE
12.2

Database Systems: Levels of Data and Process Distribution

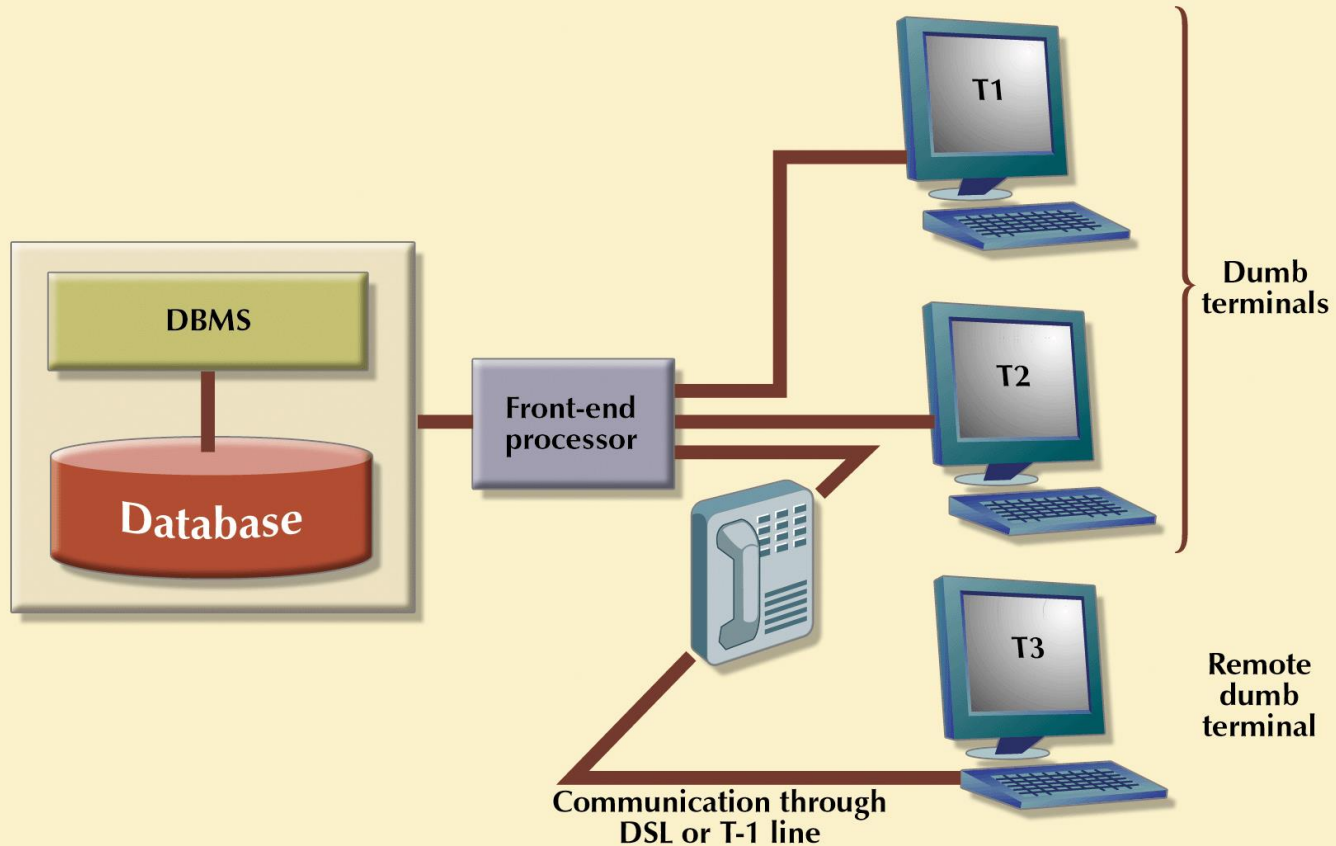
	SINGLE-SITE DATA	MULTIPLE-SITE DATA
Single-site process	Host DBMS	Not applicable (Requires multiple processes)
Multiple-site process	File server Client/server DBMS (LAN DBMS)	Fully distributed Client/server DDBMS

Single-Site Processing, Single-Site Data (SPSD)

- All processing is done on single CPU or host computer (mainframe, midrange, or PC)
- All data are stored on host computer's local disk
- Processing cannot be done on end user's side of system
- Typical of most mainframe and midrange computer DBMSs
- DBMS is located on host computer, which is accessed by dumb terminals connected to it

**FIGURE
12.6**

Single-site processing, single-site data (centralized)

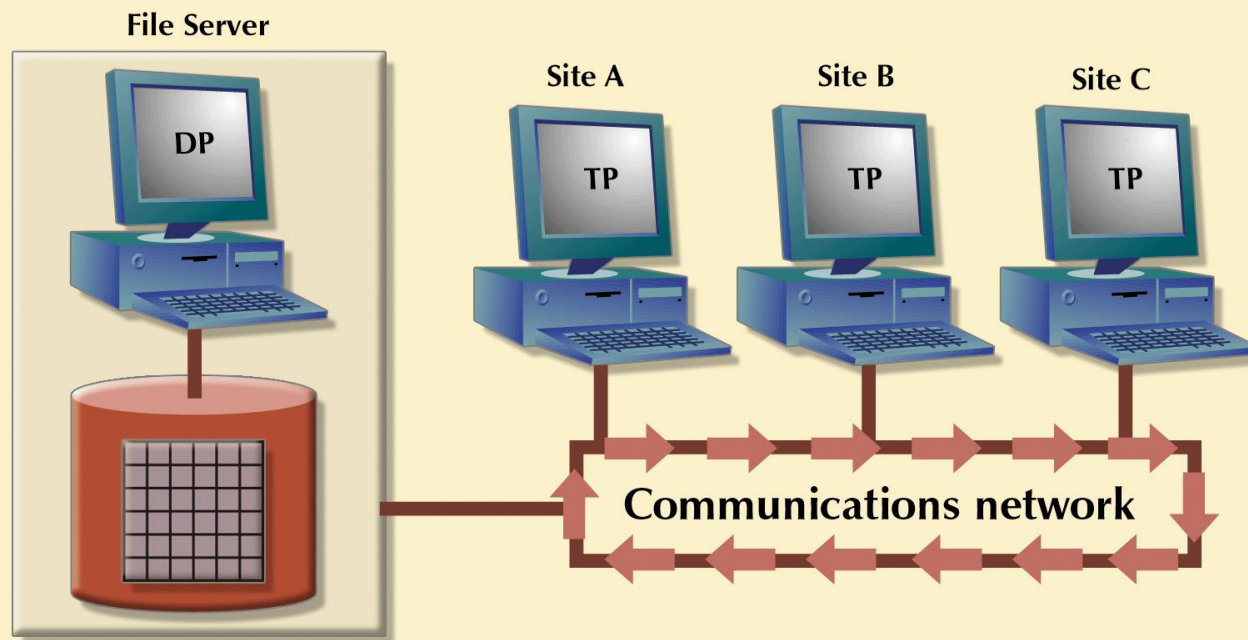


Multiple-Site Processing, Single-Site Data (MPSD)

- Multiple processes run on different computers sharing single data repository
- MPSD scenario requires network file server running conventional applications
 - Accessed through LAN
- Many multiuser accounting applications, running under personal computer network

**FIGURE
12.7**

Multiple-site processing, single-site data



Multiple-Site Processing, Multiple-Site Data (MPMD)

- Fully distributed database management system
- Support for multiple data processors and transaction processors at multiple sites
- Classified as either homogeneous or heterogeneous
- **Homogeneous DDBMSs**
 - Integrate only one type of centralized DBMS over a network

Multiple-Site Processing, Multiple-Site Data (MPMD) (continued)

- **Heterogeneous DDBMSs**
 - Integrate different types of centralized DBMSs over a network
- **Fully heterogeneous DDBMSs**
 - Support different DBMSs
 - Support different data models (relational, hierarchical, or network)
 - Different computer systems, such as mainframes and microcomputers

**FIGURE
12.8**

Heterogeneous distributed database scenario

	Platform	DBMS	Operating System	Network Communications Protocol
	IBM 3090	DB2	MVS	APPC LU 6.2
	DEC/VAX	VAX rdb	OpenVMS	DECnet
	IBM AS/400	SQL/400	OS/400	3270
	RISC computer	Informix	UNIX	TCP/IP
	Pentium CPU	Oracle	Windows Server 2003	TCP/IP

Distributed Database Transparency Features

- Allow end user to feel like database's only user
- Features include:
 - **Distribution transparency**
 - **Transaction transparency**
 - **Failure transparency**
 - **Performance transparency**
 - **Heterogeneity transparency**

Distribution Transparency

- Allows management of physically dispersed database as if centralized
- Three levels of distribution transparency:
 - **Fragmentation transparency**
 - **Location transparency**
 - **Local mapping transparency**

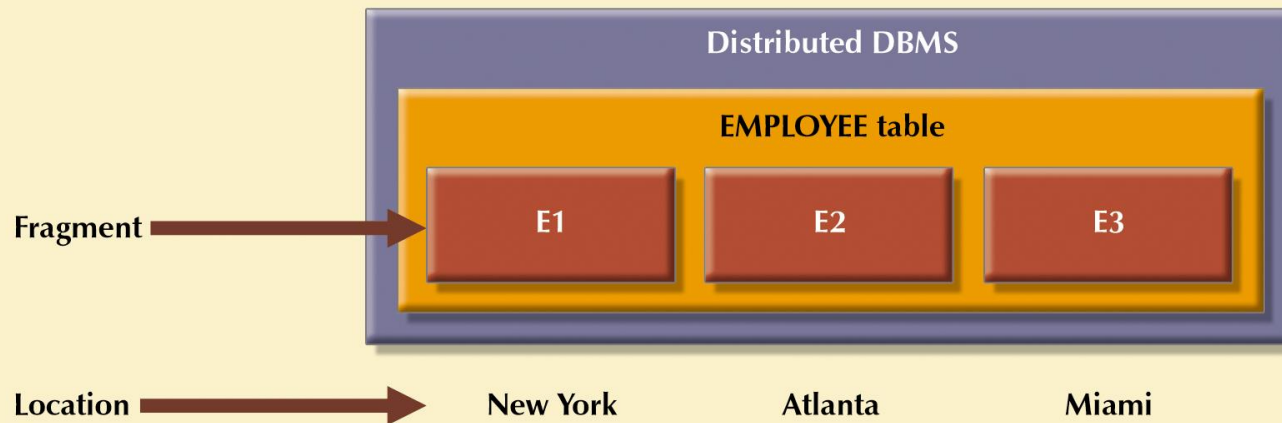
**TABLE
12.3**

A Summary of Transparency Features

IF THE SQL STATEMENT REQUIRES:			
FRAGMENT NAME?	LOCATION NAME?	THEN THE DBMS SUPPORTS	LEVEL OF DISTRIBUTION TRANSPARENCY
Yes	Yes	Local mapping	Low
Yes	No	Location transparency	Medium
No	No	Fragmentation transparency	High

**FIGURE
12.9**

Fragment locations



Transaction Transparency

- Ensures database transactions will maintain distributed database's integrity and consistency
- Ensures transaction completed only when all database sites involved complete their part
- Distributed database systems require complex mechanisms to manage transactions
 - To ensure consistency and integrity

Distributed Requests and Distributed Transactions

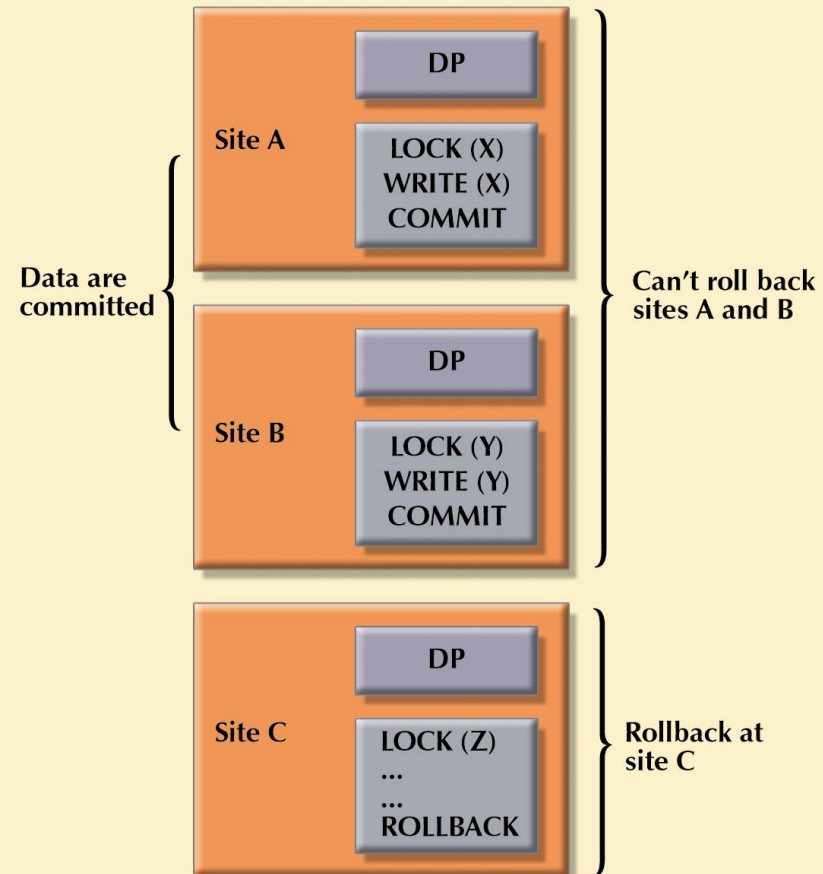
- **Remote request:** single SQL statement accesses data from single remote database
- **Remote transaction:** accesses data at single remote site
- **Distributed transaction:** requests data from several different remote sites on network
- **Distributed request:** single SQL statement references data at several DP sites

Distributed Concurrency Control

- Concurrency control important in distributed environment
 - Multisite multiple-process operations create inconsistencies and deadlocked transactions

**FIGURE
12.15**

The effect of a premature COMMIT



Two-Phase Commit Protocol

- Distributed databases make it possible for transaction to access data at several sites
- Final COMMIT issued after all sites have committed their parts of transaction
- Requires each DP's transaction log entry be written before database fragment updated
- DO-UNDO-REDO protocol with write-ahead protocol
- Defines operations between **coordinator** and **subordinates**

Performance Transparency and Query Optimization

- Query optimization routine minimizes total cost of request
- Costs a function of:
 - Access time (I/O) cost
 - Communication cost
 - CPU time cost
- Must provide distribution transparency as well as replica transparency

Performance Transparency and Query Optimization (continued)

- **Replica transparency**
 - DDBMS's ability to hide existence of multiple copies of data from user
- Query optimization:
 - Manual or automatic
 - Static or dynamic
 - Statistically based or rule-based algorithms

Distributed Database Design

- Data fragmentation
 - How to partition database into fragments
- Data replication
 - Which fragments to replicate
- Data allocation
 - Where to locate those fragments and replicas

Data Fragmentation

- Breaks single object into two or more segments or fragments
- Each fragment can be stored at any site over computer network
- Information stored in distributed data catalog (DDC)
 - Accessed by TP to process user requests

Data Fragmentation (continued)

- Strategies
 - **Horizontal fragmentation**
 - Division of a relation into subsets (fragments) of tuples (rows)
 - **Vertical fragmentation**
 - Division of a relation into attribute (column) subsets
 - **Mixed fragmentation**
 - Combination of horizontal and vertical strategies

Data Replication

- Data copies stored at multiple sites served by computer network
- Fragment copies stored at several sites to serve specific information requirements
 - Enhance data availability and response time
 - Reduce communication and total query costs
- **Mutual consistency rule:** all copies of data fragments must be identical

Data Replication (continued)

- **Fully replicated database**
 - Stores multiple copies of each database fragment at multiple sites
 - Can be impractical due to amount of overhead
- **Partially replicated database**
 - Stores multiple copies of some database fragments at multiple sites
- **Unreplicated database**
 - Stores each database fragment at single site
 - No duplicate database fragments

Data Allocation

- Deciding where to locate data
 - **Centralized data allocation**
 - Entire database is stored at one site
 - **Partitioned data allocation**
 - Database is divided into several disjointed parts (fragments) and stored at several sites
 - **Replicated data allocation**
 - Copies of one or more database fragments are stored at several sites

Client/Server vs. DDBMS

- Way in which computers interact to form system
- Features user of resources, or client, and provider of resources, or server
- Can be used to implement a DBMS in which client is the TP and server is the DP

Client/Server vs. DDBMS (continued)

- Client/server advantages
 - Less expensive than alternate minicomputer or mainframe solutions
 - Allows end user to use microcomputer's GUI, thereby improving functionality and simplicity
 - More people in job market have PC skills than mainframe skills
 - PC is well established in workplace

Client/Server vs. DDBMS (continued)

- Client/server advantages (continued)
 - Data analysis and query tools facilitate interaction with DBMSs
 - Considerable cost advantage to offloading applications development to PCs

Client/Server vs. DDBMS (continued)

- Client/server disadvantages
 - More complex environment
 - Increase in number of users and processing sites causes security problems
 - Possible to spread data access to much wider circle of users
 - Increases demand for people with broad knowledge of computers and software
 - Increases burden of training and cost of maintaining the environment

C. J. Date's Twelve Commandments for Distributed Databases

- Local site independence
- Central site independence
- Failure independence
- Location transparency
- Fragmentation transparency
- Replication transparency

C. J. Date's Twelve Commandments for Distributed Databases (continued)

- Distributed query processing
- Distributed transaction processing
- Hardware independence
- Operating system independence
- Network independence
- Database independence

Summary

- Distributed database: logically related data in two or more physically independent sites
 - Connected via computer network
- Distributed processing: division of logical database processing among network nodes
- Distributed databases require distributed processing
- Main components of DDBMS are transaction processor and data processor

Summary (continued)

- Current database systems:
 - Homogeneous distributed database system
 - Integrates one type of DBMS over computer network
 - Heterogeneous distributed database system
 - Integrates several types of DBMS over computer network

Summary (continued)

- DDBMS characteristics are a set of transparencies
- Transaction is formed by one or more database requests
- Distributed concurrency control is required in network of distributed databases
- Distributed DBMS evaluates every data request
 - Finds optimum access path in distributed database

Summary (continued)

- The design of distributed database must consider fragmentation and replication of data
- Database can be replicated over several different sites on computer network
- Client/server architecture: two computers interact over a network to form a system